

IMPACT HEALTH SYSTEMS BLUEPRINT

Powered by True North Operating System

Current 
YOUR 12-WEEK SYSTEMS BLUEPRINT

Labs show what's active right now across your core systems. Your DNA adds context so we can build a 12-week plan that fits your body, not a generic template.

SYSTEM FOCUS (12-WEEK GOALS)

PRIMARY BIOMARKER

Glucose & Insulin Control (Fasting Insulin)
8.92 µIU/mL → Goal: 2–5 µIU/mL (optimal)

SECONDARY SYSTEM FOCUS

Iron Status (Ferritin)
21.7 ng/mL → Goal: 50–150 ng/mL (optimal)

SUPPORTING SYSTEMS

Lipoprotein & Cardiovascular Risk (ApoB)
81 mg/dL Goal: < 60 mg/dL (optimal)

YOUR SYSTEMS & GENETICS

NUTRITION ARCHETYPE

Carb Controller

Stabilizes blood sugar and insulin so your body can shift into fat-burning mode.

KEY GENETIC MARKERS

System	Genes	Pattern
Systemic Inflammation	CRP, IL-10	CC, TT
Glucose & Insulin Control	FTO, SLC2A2	AA, AG
Lipoprotein Risk	APOE (ε3/ε4), LDLR	CT, TC
Foundational Support	VDR, DHCR7	TC, TT

YOUR CORE SUPPORT

SUPPLEMENT DASHBOARD

Supplement	Daily Dose
Omega-3s	2–3 g EPA/DHA
Berberine	500 mg 2–3×
Methylated B	1 cap
Soluble Fiber	7–10 g
Iron	Per provider

Methylation	MTHFR A1298C	TG

HOW WE USE YOUR LABS & DNA

Labs show what's active right now and set the focus for this 12-week block across five key systems: **systemic inflammation, glucose & insulin control, lipoprotein & cardiovascular risk, methylation & cellular repair, and foundational support (vitamin D and related factors)**. Your DNA adds context, helping explain why certain markers look the way they do and how to fine-tune your nutrition, supplements, and recovery. Together, they create a plan that's built for your body, not a generic template.

This report distinguishes between clinical ranges (disease detection) and optimal ranges (performance and long-term health). Recommendations are based on optimizing physiology, not just avoiding disease.

PRIMARY SYSTEM FOCUS: GLUCOSE & INSULIN CONTROL (FASTING INSULIN)

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Current (Fasting Insulin): Current: 8.92 µIU/mL
Target (Fasting Insulin): 2–5 µIU/mL (optimal)

Why It Matters

Insulin is a key metabolic lever. When fasting insulin is elevated, your body is stuck in storage mode — making it harder to burn fat, regulate appetite, and sustain energy between meals. Over time, elevated insulin drives insulin resistance, which compounds metabolic dysfunction and increases the risk of type 2 diabetes, cardiovascular disease, and chronic inflammation. We address it now because insulin is active — it's the main focus of this block.

What Happens If We Don't

Elevated insulin quietly works against you. Fat loss stalls, hunger becomes harder to manage, and energy fluctuates throughout the day. Over time, the pancreas works harder to compensate, and the downstream effects — weight gain, increased inflammation, hormonal disruption — become harder to reverse. We address this early because the longer insulin stays elevated, the more entrenched the metabolic pattern becomes.

How We're Addressing It

Nutrition structure designed to reduce insulin demand — emphasizing protein, fiber, and healthy fats while managing carbohydrate timing and quality. Berberine to directly improve insulin sensitivity at the cellular level. Movement and meal timing strategies to support glucose clearance without relying on more insulin.

Genetic Context: Glucose & Insulin Control (Fasting Insulin)

Your genetic pattern shapes how your body manages insulin signaling, glucose uptake, and metabolic flexibility. These variants influence how efficiently your cells respond to insulin, how your pancreas regulates secretion, and how your appetite and energy balance systems interact. The genes below help explain where those sensitivities may exist and why glucose and insulin optimization is being prioritized now.

Gene	RS Number	Pathway Role	Your Result
TCF7L2	rs7903146	Beta-cell function & insulin secretion	CC

FTO	rs9939609	Energy balance & appetite regulation	AA
MC4R	rs17782313	Central appetite signaling	TT
SLC2A2	rs5400	Glucose transporter expression	AG
PPARG	rs1801282	Insulin sensitivity & fat metabolism	CG
MTNR1B	rs10830963	Melatonin receptor & glucose metabolism	CG

SLC2A2 (GLUT2) (rs5400) | Effect Allele: rs5400 = A

SLC2A2 encodes the GLUT2 glucose transporter responsible for hepatic glucose sensing. Your pattern is associated with a tendency toward impaired glucose detection by the liver, which can delay the appropriate insulin response and contribute to post-meal glucose spikes.

PPARG (rs1801282) | Effect Allele: rs1801282 = C

PPARG regulates fat cell differentiation and peripheral insulin sensitivity. Your pattern is associated with a tendency toward reduced insulin sensitivity in adipose tissue, meaning your fat cells are less responsive to insulin signaling — which can shift glucose handling toward the liver and muscle, increasing overall metabolic demand.

FTO (rs9939609) | Effect Allele: rs9939609 = A

FTO influences energy intake regulation and metabolic efficiency. Your pattern is associated with a tendency toward increased hunger signaling and reduced post-meal satiety, which can drive higher caloric intake and place additional demand on your glucose-insulin system.

MTNR1B (rs10830963) | Effect Allele: rs10830963 = G

Your genetic pattern in MTNR1B is associated with a tendency toward elevated fasting glucose linked to melatonin signaling. This gene encodes the melatonin receptor expressed on pancreatic beta cells, and this variation can mean melatonin has a stronger-than-average suppressive effect on insulin secretion — which is why fasting glucose may be higher in the morning and why meal timing relative to sleep can influence your glucose control.

GCK (rs1799884) | Effect Allele: rs1799884 = T

GCK encodes glucokinase, one of the primary enzymes responsible for sensing and processing glucose in the liver and pancreas. Your genetic pattern is associated with a tendency toward a higher threshold for glucose sensing, meaning insulin response may not be triggered as efficiently until blood sugar is slightly higher. In your pattern, this helps explain why fasting glucose or post-meal glucose handling may drift upward more easily under stress, disrupted sleep, or inconsistent nutrition, reinforcing the plan's focus on steady glucose control and metabolic consistency.

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SECONDARY SYSTEM FOCUS: IRON STATUS (FERRITIN)

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Current (Ferritin): Current: 21.7 ng/mL
Target (Ferritin): 50–150 ng/mL (optimal)

Why It Matters

Ferritin reflects your body's iron storage. While it's not the primary focus this block, iron that's out of range — too low or too high — quietly undermines oxygen delivery, energy production, and recovery capacity.

What Happens If We Don't

If we address the primary target but ignore iron status, fatigue and poor recovery may persist regardless of other improvements. Iron imbalances compound over time and make every other intervention less effective.

How We're Addressing It

Iron management is included based on your current levels and genetic risk profile. The trend is worth rechecking at 12 weeks to confirm response and adjust strategy.

Genetic Context: Iron Status (Ferritin)

Your genetic pattern influences how your body absorbs, transports, and stores iron. These variants can explain why some people accumulate iron more readily or struggle with iron utilization. The genes below help explain where those sensitivities may exist and why iron status management is being prioritized now.

Gene	RS Number	Pathway Role	Your Result
HFE	rs1800562, rs1799945	Iron absorption regulation	GG
TF	rs3811647	Iron transport protein	AG

TF (rs3811647) | Effect Allele: rs3811647 = G

Your genetic pattern in TF is associated with a tendency toward altered transferrin levels. Transferrin is the primary iron transport protein in the blood, and this variation can affect how efficiently iron is delivered to tissues — influencing both ferritin storage levels and functional iron availability.

SUPPORTING SYSTEM: CARDIOVASCULAR RISK

Current (ApoB): Current: 81 mg/dL
Target (ApoB): < 60 mg/dL (optimal)

Why It Matters

ApoB counts the number of atherogenic particles in your blood — the vehicles that deposit cholesterol into arterial walls. It's a more accurate predictor of cardiovascular risk than LDL cholesterol alone. Even when other markers look normal, elevated ApoB means your vascular system is under pressure. We address it now because cardiovascular risk compounds silently, and ApoB responds well to targeted nutrition and lifestyle intervention.

What Happens If We Don't

Elevated ApoB is silent — you won't feel it until significant damage is done. Plaque accumulates slowly but relentlessly, narrowing arteries and increasing the risk of heart attack and stroke. Unlike inflammation or insulin, there are no obvious symptoms to alert you. We address this now because the window for prevention is wide open, and the interventions are straightforward.

How We're Addressing It

Targeted nutrition to reduce atherogenic particle production — emphasizing soluble fiber to increase LDL receptor activity, omega-3s to improve lipid metabolism, and dietary fat quality optimization. Your genetic profile informs how aggressively we manage saturated fat intake and whether additional interventions may be warranted.

Genetic Context: Lipoprotein & Cardiovascular Risk (ApoB)

ApoB reflects the number of atherogenic lipoprotein particles circulating in the blood. Particle burden is shaped by how many particles are produced, how efficiently they are cleared, and how strongly dietary inputs influence that balance.

Gene	RS Number	Pathway Role	Your Result
APOE	rs429358 + rs7412	Lipoprotein clearance & remnant handling	CT
LDLR	rs688	LDL receptor recognition & particle uptake	TC
APOB	rs693	ApoB particle assembly & production	AG
CETP	rs708272	Lipid exchange between lipoproteins	AG
LPA	rs3798220, rs10455872	Lipoprotein(a) particle burden	TT
PCSK9	rs11591147	LDL clearance regulation	GG
HMGCR	rs17238540	Statin pathway / cholesterol synth	TT
ANGPTL3	rs1748195	Triglyceride regulation	CG
APOB2	rs1042031	ApoB particle production	CC

SORT1	rs646776	LDL production (secondary)	TT
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APOE (rs429358, rs7412) | Effect Allele: rs429358 = C, rs7412 = C

APOE plays a central role in lipoprotein clearance and remnant handling. Your pattern is associated with a tendency toward less efficient clearance of LDL and remnant lipoprotein particles from the bloodstream, which contributes to a higher baseline ApoB level and increased cardiovascular risk — particularly in the context of saturated fat intake.

LDLR (rs688) | Effect Allele: rs688 = T

LDLR encodes the LDL receptor responsible for pulling LDL particles out of the bloodstream and into the liver for processing. Your pattern is associated with a tendency toward reduced receptor density or efficiency, meaning your liver clears LDL particles more slowly — contributing to elevated ApoB and LDL-C levels.

APOB (rs693) | Effect Allele: rs693 = G

APOB encodes the apolipoprotein B protein — the structural backbone of every LDL particle. Your pattern is associated with a tendency toward higher ApoB particle production from the liver, which directly increases the number of atherogenic particles in circulation regardless of their cholesterol content.

CETP (rs708272) | Effect Allele: rs708272 = A

CETP mediates the exchange of lipids between HDL and LDL particles. Your pattern is associated with a tendency toward increased CETP activity, which shifts cholesterol from protective HDL particles to atherogenic LDL particles — contributing to a less favorable lipid profile even when total cholesterol appears normal.

PCSK9 (rs11591147) | Effect Allele: rs11591147 = G

Your genetic pattern in PCSK9 is associated with a tendency toward higher circulating LDL levels. This gene regulates how many LDL receptors are available on the surface of liver cells to clear LDL from the bloodstream, and this variation can mean your liver is less efficient at removing LDL particles — contributing to a higher baseline ApoB level independent of diet.

SORT1 (rs646776) | Effect Allele: rs646776 = T

Your genetic pattern in SORT1 is associated with a tendency toward increased hepatic LDL secretion. This gene influences how the liver packages and releases lipoprotein particles into the bloodstream, and this variation can mean your system produces more LDL particles than average — which helps explain why ApoB may remain elevated even when dietary inputs are well-controlled.

HMGCR (rs17238540) | Effect Allele: rs17238540 = T

Your genetic pattern in HMGCR is associated with a tendency toward higher endogenous cholesterol production. This gene encodes the rate-limiting enzyme in cholesterol synthesis — the same enzyme targeted by statin medications — and this variation can mean your liver produces more cholesterol internally, which contributes to elevated LDL and ApoB levels beyond what dietary cholesterol intake alone would explain.

CETP Transfer (rs5882) | Effect Allele: rs5882 = G

CETP helps transfer cholesterol between different lipoproteins in the bloodstream, influencing how cholesterol is redistributed across particles. Your genetic pattern is associated with altered transfer activity, which can shift how cholesterol is packaged between HDL and apoB-containing particles. In your pattern, this

adds context to lipid handling — specifically how particles are structured and redistributed — but does not change the primary driver of risk, which is the total number of apoB-containing particles measured by ApoB.

CLINICAL ACTION SUMMARY

Your elevated ApoB level indicates a higher than optimal number of atherogenic particles in your bloodstream, which is a primary driver of cardiovascular risk. Your genetic profile provides important context, suggesting a tendency toward less efficient clearance and/or increased production of these particles, which can raise your baseline risk when diet and lifestyle are not dialed in. The following strategies are designed to directly address these underlying mechanisms.

- **Manage Saturated Fat Intake:** Your genetic risk profile may make your system more sensitive to the LDL-raising effects of saturated fat. This type of fat can down-regulate the LDL receptors on your liver, reducing its ability to clear ApoB-containing particles. Prioritize healthy fats from whole-food sources like avocados, olive oil, nuts, and seeds, while being mindful of your intake from red meat, full-fat dairy, and coconut oil.
- **Increase Soluble Fiber:** Soluble fiber is a powerful tool for lowering ApoB. It binds to bile acids in the gut, forcing your liver to pull more cholesterol from the bloodstream to produce new bile acids. This directly increases LDL receptor activity. Aim for at least 10 g/day of soluble fiber (and more if tolerated) from sources like oats, barley, psyllium husk, apples, and beans.
- **Consider Zone 2 Cardio:** While all exercise is beneficial, steady-state cardiovascular exercise (often called Zone 2) is particularly effective at improving mitochondrial function and the efficiency with which your body uses fat for fuel. This can improve lipid metabolism and, alongside diet, contribute to lower levels of circulating atherogenic particles over time. Incorporating regular sessions of 45-60 minutes at a conversational pace is a sustainable way to support your cardiovascular health.
- **Consider Plant Sterols:** Foods fortified with plant sterols and stanols can be an effective adjunct strategy. They work by blocking the absorption of cholesterol in the gut, which typically produces a modest (roughly 5-10%) reduction in LDL cholesterol and ApoB when used consistently with other dietary changes.

These genetic patterns do not determine cardiovascular risk on their own. They help explain why ApoB may rise more easily or respond strongly to dietary inputs, particularly fat quality and fiber availability.

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LIFESTYLE, STRESS & RECOVERY

This section was triggered because you indicated that recovery limitations are a frequent concern. Your genetic profile reveals specific tendencies in how your body manages stress, processes stimulants like caffeine, and regulates your internal clock — all of which directly impact your ability to recover.

Why It Matters

Your recovery capacity is not just about rest — it's a function of how your nervous system, circadian rhythm, and stress-response pathways operate at the genetic level. Variants in genes like COMT, CYP1A2, PER3, and FKBP5 can influence everything from how quickly you clear stress hormones to how efficiently you metabolize caffeine and when your body naturally wants to sleep. Understanding these patterns allows us to build a recovery strategy that works with your biology, not against it.

What Happens If We Don't

When your recovery systems are misaligned — whether from poor sleep timing, excessive stimulant use, or a stress response that doesn't downregulate efficiently — the effects cascade across every other system in your body. Inflammation increases, insulin sensitivity decreases, and your body shifts from a state of repair to a state of survival. Over time, this creates a compounding deficit that undermines training, body composition, and cognitive performance.

How We're Addressing It

Your protocol includes a targeted recovery strategy. This involves optimizing your sleep hygiene based on your circadian tendencies, managing caffeine intake according to your metabolic profile, and implementing targeted stress-reduction protocols that align with your neurological and cortisol-related genes. The goal is not to eliminate stress, but to build a robust recovery infrastructure that allows your body to adapt and thrive. This includes specific recommendations for sleep timing, light exposure, and daily decompression practices that are foundational to your success.

Gene	RS Number	Pathway Role	Your Result
COMT	rs4680	Dopamine clearance speed	GG
OXTR	rs53576	Stress resilience	GG
OPRM1	rs1799971	Pain/reward signaling	AA
BDNF	rs6265	Neuroplasticity & recovery	CC
CYP1A2	rs762551	Caffeine metabolism speed	AA
AHR	rs4410790	Caffeine sensitivity / receptor	CC
CYP2A6	rs1801272	Caffeine / nicotine metabolism	AA
CLOCK	rs1801260	Circadian rhythm regulation	AA
CRY1	rs2287161	Delayed sleep phase	CC
ADA	rs73598374	Sleep pressure / adenosine	CC
MTNR1B	rs10830963	Melatonin receptor sensitivity	CG
FKBP5	rs1360780	Cortisol sensitivity / recovery	--
ADRB2	rs1042713	Adrenergic stress response	GG
PPARGC1A	rs8192678	Mitochondrial biogenesis	TC

Neurotransmitter & Stress Recovery

These genes influence dopamine clearance, oxytocin signaling, opioid receptor sensitivity, and neuroplasticity — all of which shape how your nervous system responds to and recovers from stress.

Caffeine Metabolism

These genes determine how quickly you metabolize caffeine and how sensitive your receptors are to its effects. This influences optimal timing, dose, and whether caffeine supports or disrupts your performance and sleep.

CYP1A2 (rs762551) | Effect Allele: rs762551 = A

Your genetic pattern in CYP1A2 is associated with slower caffeine metabolism. This gene encodes the liver enzyme responsible for breaking down roughly 95% of ingested caffeine, and this variation means caffeine stays active in your system significantly longer — amplifying both its performance benefits and its potential to disrupt sleep, elevate cortisol, and increase heart rate if consumed too late in the day.

CYP2A6 (rs1801272) | Effect Allele: rs1801272 = A

Your genetic pattern in CYP2A6 is associated with a tendency toward altered secondary caffeine metabolism. While CYP1A2 handles the primary breakdown, CYP2A6 processes several caffeine metabolites, and this variation can mean those downstream products linger longer — contributing to prolonged stimulant effects and a wider window of sleep disruption.

Circadian Rhythm & Sleep

These genes influence your internal clock, sleep pressure, and melatonin signaling. Variants here can explain why some people are natural night owls, struggle with sleep onset, or are more sensitive to light disruption.

CLOCK (rs1801260) | Effect Allele: rs1801260 = A

Your genetic pattern in CLOCK is associated with a tendency toward a delayed circadian phase and potential difficulty with sleep onset. This gene is the master regulator of your circadian rhythm, and this variation can mean your internal clock runs slightly longer than 24 hours — contributing to a natural drift toward later bedtimes and morning grogginess.

MTNR1B (rs10830963) | Effect Allele: rs10830963 = G

Your genetic pattern in the melatonin receptor gene MTNR1B is associated with a stronger-than-average impact of melatonin on your circadian rhythm. While this can be beneficial for sleep, it also means melatonin has a more pronounced suppressive effect on insulin secretion — linking your sleep-wake cycle directly to glucose regulation.

Stress & Cortisol Response

These genes influence how your HPA axis responds to stress, how quickly cortisol clears, and how your adrenergic and serotonin systems modulate the stress response. Variants here can explain prolonged recovery from high-stress periods.

NR3C1 (rs6190) | Effect Allele: rs6190 = C

Your genetic pattern in NR3C1 is associated with a tendency toward altered glucocorticoid receptor sensitivity. This gene encodes the receptor that cortisol binds to in nearly every tissue in your body, and this variation can mean your cells are less responsive to cortisol's regulatory signals — potentially requiring higher cortisol levels to achieve the same stress-dampening effect.

ADRB2 (rs1042713) | Effect Allele: rs1042713 = G

Your genetic pattern in the beta-2 adrenergic receptor is associated with a tendency toward higher adrenaline and noradrenaline signaling. These are your "fight or flight" hormones, and this variation can mean your system mounts a stronger sympathetic response to stress — useful for acute performance but problematic when chronically activated.

Mitochondrial & Cellular Energy

These genes influence mitochondrial biogenesis and cellular energy regulation. Variants here affect how efficiently your cells produce energy and adapt to metabolic demands.

PPARGC1A (rs8192678) | Effect Allele: rs8192678 = T

Your genetic pattern here is associated with a tendency toward a slower cellular rebuilding process following physical or mental stress — meaning your system may need more time between high-demand periods to fully restore.

CLINICAL ACTION SUMMARY

This section is triggered because you indicated that recovery limitations are a frequent concern, and your genetic profile provides context for why that might be. Your genes suggest tendencies related to your internal stress response system (the HPA axis), your natural sleep-wake cycle (circadian rhythm), and the metabolism of key neurotransmitters and compounds like caffeine. The following strategies are designed to help you build a more robust recovery system.

Lean Into Your Chronotype: Your genetic profile suggests tendencies that may lean you toward being more of a night owl or morning lark. Instead of fighting this, try to structure your day to align with your natural energy patterns. If you tend to be a night owl, consider scheduling your most demanding tasks for later in the day. Regardless of your chronotype, consistent light exposure in the morning is a powerful tool to anchor your daily rhythm.

Implement a Proactive Stress Off-Ramp: Your genetic profile suggests your system may be slower to turn off the cortisol response after a stressor. Implementing a daily 10-20 minute off-ramp practice such as non-sleep deep rest (NSDR), breathwork, or meditation are highly effective tools for signaling safety to your body and helping your HPA axis down-regulate.

Respect Your Caffeine Metabolism: Your genetic profile suggests a tendency toward slower caffeine metabolism. This means caffeine may stay in your system significantly longer. Consider testing a strict caffeine cutoff time (e.g., 12 PM or earlier) and paying close attention to how it affects your sleep quality and feelings of anxiety or restlessness.

Manage Your Dopamine Environment: If your genetic profile includes a certain COMT variation, it suggests your brain may be slower to clear dopamine. This can support focus but may also increase susceptibility to overstimulation or anxiety. Be mindful of stacking stimulants and consider scheduling regular breaks from demanding cognitive work to allow your neurotransmitter systems to reset.

Carb Controller

Stabilize Blood Sugar, Restore Metabolic Flexibility

Your nutrition archetype is: Carb Controller. This approach improves insulin sensitivity so calories are directed toward muscle instead of fat. When insulin signaling is impaired, hunger rises, energy crashes, and muscle is harder to retain — even in a calorie deficit. By controlling carbohydrate dose, timing, and quality, this plan restores metabolic efficiency and makes progress more sustainable.

This Diet Revolves Around

Structured carbohydrate intake, higher protein emphasis, controlled portions, and strategic carb timing around training. Carbs are not eliminated — they're used intentionally.

Why This Fits Your Body

This archetype matches your insulin and glucose pattern — and, where present, genes like TCF7L2, PPARG, and SLC2A2. The biomarker section explains why those matter; this plan is how you make them work in your favor.

Cardiovascular Optimization Note

Because your ApoB is elevated, we're also emphasizing fat quality across this plan — limiting saturated fats (fatty cuts of meat, butter, cheese, coconut oil) and prioritizing monounsaturated and omega-3 sources (olive oil, avocado, nuts, fatty fish). This layers on top of your nutrition archetype, not instead of it.

YOUR SUPPLEMENT PROTOCOL

Supplement	Daily Dose	When To Take	Primary Purpose
Omega-3s (EPA/DHA)	2-3g combined EPA/DHA daily	With meals, can split AM/PM	Anti-inflammatory, cardiovascular, recovery
Berberine	500mg 2-3x daily with meals	With meals, titrate up from 500mg 1x daily	Insulin sensitivity, metabolic support
Methylated B-Complex	1 capsule daily (methylfolate + methylcobalamin)	Morning with food	Methylation support, homocysteine clearance
Soluble Fiber	7-10g/day (6-8g if gut-sensitive)	Split across meals, titrate from 3-4g/day	ApoB reduction, gut health, blood sugar
Iron	Per provider recommendation	On empty stomach or with vitamin C	Iron repletion

Omega-3s (EPA/DHA)

Dose: 2-3g combined EPA/DHA daily

Timing: With meals, can split AM/PM

Why It's Included

Omega-3 fatty acids are foundational for managing inflammation, supporting cardiovascular health, and optimizing cell membrane function. EPA and DHA modulate inflammatory signaling at the cellular level, improve lipid profiles, and support recovery from training stress.

Practical Notes

Look for supplements with at least 1g EPA + 1g DHA combined per serving. Triglyceride form is better absorbed than ethyl ester form. Can cause fishy burps — taking with meals and freezing capsules helps. If vegetarian/vegan, algae-based EPA/DHA is an equivalent alternative.

Berberine

Dose: 500mg 2-3x daily with meals

Timing: With meals, titrate up from 500mg 1x daily

Why It's Included

Berberine is a plant compound that activates AMPK — the same metabolic switch triggered by exercise and caloric restriction. It has been shown in research to meaningfully improve insulin sensitivity. For individuals with elevated fasting insulin or glucose, berberine addresses the root dysfunction.

Practical Notes

Start low and titrate — GI upset is common if starting at full dose. May lower blood sugar — if on diabetes medications, coordinate with physician. Can take 2-4 weeks to see full effect on fasting insulin.

Methylated B-Complex

Dose: 1 capsule daily (methylfolate + methylcobalamin)

Timing: Morning with food

Why It's Included

Methylated B-vitamins support the methylation cycle — a foundational process for DNA repair, neurotransmitter production, homocysteine clearance, and cellular energy. Methylated forms are "pre-activated" and bypass genetic limitations that can make standard B-vitamins less effective.

Practical Notes

Look for products listing "methylfolate" (or 5-MTHF) and "methylcobalamin" — not "folic acid" or "cyanocobalamin". Folate-rich foods (leafy greens, legumes) complement supplementation. Worth rechecking homocysteine at 12 weeks.

Soluble Fiber

Dose: 7-10g/day (6-8g if gut-sensitive)

Timing: Split across meals, titrate from 3-4g/day

Why It's Included

Soluble fiber binds bile acids in the gut, which forces the liver to pull cholesterol from the bloodstream to make more bile. This mechanism directly supports LDL particle clearance and ApoB reduction. It also supports gut health, blood sugar stability, and satiety.

Practical Notes

Titrate slowly — jumping to full dose causes bloating and gas. Take with plenty of water. Common sources: psyllium husk, PHGG, oat beta-glucan, apple pectin. Best split across meals rather than taken all at once.

PEPTIDE CONSIDERATIONS

Important

The peptides listed below are sometimes discussed with medical providers as adjuncts to nutrition, training, and lifestyle optimization. This section is informational only — peptides require provider evaluation, prescription, and monitoring. Nothing here constitutes a recommendation or prescription. Discuss options with a qualified provider if interested.

Metabolic Support Triggered by elevated insulin/glucose or aggressive fat-loss goal

Semaglutide

A GLP-1 agonist commonly discussed for appetite regulation and glucose signaling. Sometimes considered to support adherence to calorie targets and metabolic markers when fat-loss goals are aggressive.

Tirzepatide

A dual-agonist compound (GLP-1 and GIP) discussed for supporting appetite regulation and metabolic signaling. Sometimes considered when providers want a broader metabolic approach.

Retatrutide (Advanced)

A triple-agonist compound (GLP-1, GIP, and glucagon) representing a newer approach to metabolic support. Typically only discussed in provider-led contexts.

MOTS-C

A mitochondrial-derived peptide discussed in research contexts for its potential role in cellular energy metabolism and metabolic flexibility.

RECOVERY & NEUROLOGICAL SUPPORT Triggered by circadian, stress, or neurological genetic variants

DSIP (Delta Sleep-Inducing Peptide)

A neuropeptide that promotes deep, restorative sleep architecture by modulating delta wave activity. Supports natural sleep onset, enhances sleep quality, and improves overnight recovery. Particularly relevant for individuals with genetic variants affecting circadian rhythm or stress-related sleep disruption.

Semax / Selank

Semax is a synthetic ACTH analog that supports cognitive function, focus, and neuroprotection by enhancing BDNF expression. Selank is a tuftsin analog with anxiolytic properties that modulates GABA and serotonin pathways. Together, this combination supports stress resilience, mental clarity, and neurological recovery — especially relevant for individuals with COMT, BDNF, or serotonin transporter variants.

BLOOD WORK REFERENCE

This table summarizes the 8 biomarkers analyzed in your blueprint. Each marker is assigned a role based on your current results: Primary Focus (the main target this block), Secondary Focus (monitoring for improvement), Supporting (layered intervention), or Monitored (within range, tracked for context).

Biomarker	Your Result	Optimal Range	12-Week Target	Status
hs-CRP	Current: <0.20 mg/L	< 1.0 mg/L	< 1.0 mg/L (optimal)	Monitored
Fasting Insulin	Current: 8.92 μ IU/mL	2–5 μ IU/mL (optimal)	2–5 μ IU/mL (optimal)	Primary Focus
Fasting Glucose	79 mg/dL	70-90 mg/dL	70-90 mg/dL	Monitored
HbA1c	5.3 %	< 5.4%	< 5.4%	Monitored
Homocysteine	6.24 μ mol/L	< 9 μ mol/L	< 9 μ mol/L	Monitored
Ferritin	21.7 ng/mL	50-150 ng/mL	50-150 ng/mL	Secondary Focus
ApoB	Current: 81 mg/dL	< 60 mg/dL (optimal)	< 60 mg/dL (optimal)	Elevated
Vitamin D	Current: 65.19 ng/mL	50–70 ng/mL (optimal)	50–70 ng/mL (optimal)	Within optimal range

Status Definitions

Primary Focus — The main biomarker driving your 12-week plan. Full intervention applied.

Secondary Focus — Elevated but not the primary target. Monitored with supporting interventions.

Supporting — Addressed with targeted additions layered onto your base plan.

Monitored — Currently not a primary or secondary target. Tracked for context and rechecked at 12 weeks.

GENETIC APPENDIX

This appendix lists the 66 SNPs analyzed as part of your Impact Health Systems Blueprint. These are your raw genotype results across 10 biological categories. Where a variant is present, its context and relevance are explained in the system sections of this report. Genes without notable variants are still listed here for completeness — they confirm what's working as expected.

Lipid Handling

Gene	RS Number	What It Affects	Your Result
APOE	rs429358 + rs7412	Particle clearance, sat fat response	CT
APOB	rs693	ApoB production	AG
LDLR	rs688	LDL receptor function	TC
CETP	rs708272	HDL/LDL transfer	AG
LPA	rs3798220, rs10455872	Lp(a) levels	TT
APOE	rs7412	APOE ε2 allele (protective)	CC
LPA	rs10455872	Lp(a) levels (secondary)	AA
PPARA	rs4253778	Fatty acid oxidation	GG
—	rs4714209	LD proxy	TC
CETP	rs5882	Cholesterol transfer	AG

Inflammation

Gene	RS Number	What It Affects	Your Result
CRP	rs1205, rs1800947	Baseline CRP production	CC
IL-6	rs1800795	Pro-inflammatory signaling	GG
TNF-α	rs1800629	Inflammatory response intensity	GG
NFKB1	rs28362491	Master inflammatory switch	II
IL-10	rs1800896	Anti-inflammatory capacity	TT
CRP	rs1800947	Baseline CRP production (secondary)	--
IL-1β promoter	rs1143627	Inflammatory cytokine	AG
IL-6R	rs8192284	IL-6 receptor signaling	--

Methylation

Gene	RS Number	What It Affects	Your Result
MTHFR	rs1801133 (C677T)	Folate processing (major)	GG
MTHFR	rs1801131 (A1298C)	Folate processing (minor)	TG
MTR	rs1805087	B12-dependent remethylation	AA
MTRR	rs1801394	B12 recycling	GG
CBS	rs234706	Transsulfuration pathway	AG
BHMT	rs3733890	Alternate remethylation	GG
TCN2	rs1801198	B12 transport	CC
FUT2	rs602662	B12 absorption, gut signaling	AG

Insulin/Glucose

Gene	RS Number	What It Affects	Your Result
TCF7L2	rs7903146	Insulin production (strongest predictor)	CC
SLC2A2	rs5400	Glucose transport	AG
PPARG	rs1801282	Fat cell insulin sensitivity	CG
SLC30A8	rs13266634	Zinc transport in beta cells	CC
IRS2	rs4771647	Insulin signaling	AC
GCK	rs1799884	Glucose sensing / fasting glucose	TC

Vitamin D

Gene	RS Number	What It Affects	Your Result
CYP2R1	rs10741657	Vitamin D activation	GG
DHCR7	rs12785878	Vitamin D synthesis	TT
GC	rs2282679, rs4588, rs7041	Vitamin D binding/transport	TT
VDR	rs2228570, rs1544410, rs731236	Vitamin D receptor sensitivity	--
GC	rs4588	Vitamin D transport (secondary)	GG
GC	rs7041	Vitamin D transport (tertiary)	CC

Iron Metabolism

Gene	RS Number	What It Affects	Your Result
HFE	rs1800562 (C282Y)	Iron absorption (major)	GG

HFE	rs1799945 (H63D)	Iron absorption (minor)	CC
TF	rs3811647	Transferrin function	AG

Omega-3 Metabolism

Gene	RS Number	What It Affects	Your Result
FADS1	rs174546	ALA to EPA conversion	TC
FADS2	rs174618	EPA to DHA conversion	CC
FADS2	rs1535	EPA to DHA conversion (secondary)	AG

Gut/Digestion

Gene	RS Number	What It Affects	Your Result
LCT	rs4988235	Lactase persistence	GG
HLA-DQA1	rs2187668	Celiac susceptibility	CC
HLA-DQB1	rs7454108	Celiac susceptibility	TT
NOD2	rs2066844	Innate gut immunity	CC
AOC1 / DAO	rs10156191	Histamine breakdown	CC

Appetite/Metabolism

Gene	RS Number	What It Affects	Your Result
LEP	rs7799039	Leptin production	AG
LEPR	rs1137101	Leptin receptor	AG
MC4R	rs17782313	Appetite regulation	TT
GHRL	rs696217	Ghrelin (hunger hormone)	--
MC4R	rs6567160	Satiety regulation	TT

Connective Tissue

Gene	RS Number	What It Affects	Your Result
COL1A1	rs1800012	Collagen type I	AC
COL5A1	rs12722	Collagen type V	TC
GDF5	rs143383	Joint/cartilage development	GG
MMP3	rs679620	Tissue remodeling	TT

ACTN3	rs1815739	Muscle fiber type	CC
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Stress & Recovery

Gene	RS Number	What It Affects	Your Result
COMT	rs4680	Dopamine metabolism	GG
OXTR	rs53576	Stress resilience	GG
OPRM1	rs1799971	Reward sensitivity	AA
BDNF	rs6265	Neuroplasticity, mood	CC
GNB3	rs5443	Sympathetic signaling	CC
DRD2 / ANKK1	rs1800497	Dopamine reward	GG

Oxidative Stress

Gene	RS Number	What It Affects	Your Result
SOD2	rs4880	Mitochondrial antioxidant	GG
NQO1	rs1800566	Detoxification enzyme	AG
CAT	rs1001179	Catalase function	TC
HMOX1	rs2269533	Heme oxygenase	--
NRF2	rs6721961	Antioxidant response	--
GPX1	rs1050450	Glutathione peroxidase	AG

CAFFEINE METABOLISM

Gene	RS Number	What It Affects	Your Result
CYP1A2	rs762551	Caffeine metabolism speed	AA
AHR	rs4410790	Caffeine sensitivity / receptor	CC
CYP2A6	rs1801272	Caffeine / nicotine metabolism	AA

CIRCADIAN RHYTHM & SLEEP

Gene	RS Number	What It Affects	Your Result
CLOCK	rs1801260	Circadian rhythm regulation	AA
CRY1	rs2287161	Delayed sleep phase tendency	CC
ADA	rs73598374	Adenosine / sleep pressure	CC
MTNR1B	rs10830963	Melatonin receptor / light sensitivity	CG

STRESS & CORTISOL

Gene	RS Number	What It Affects	Your Result
FKBP5	rs1360780	Cortisol sensitivity / recovery	--
NR3C1	rs6190	Glucocorticoid receptor function	CC
ADRB2	rs1042713	Adrenergic stress response	GG

MITOCHONDRIAL FUNCTION

Gene	RS Number	What It Affects	Your Result
SIRT1	rs10997868	Metabolic regulation / longevity	--
PPARGC1A	rs8192678	Mitochondrial biogenesis	TC
UCP2	rs659366	Mitochondrial uncoupling	CC

Other

Gene	RS Number	What It Affects	Your Result
—	rs4263839	—	GG
—	rs9290264	—	CC

YOUR PLAN IS SET